Luiz Cauet Castilla Ruy Blum

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EDUCATION

University of California, Santa Cruz — Santa Cruz, California
B.S. in Computer Engineering - Systems Programming, CS Minor — *Expected Graduation, March 2027*

GPA: 3.4
Related Coursework: Machine Learning, Data Structures & Algorithms, Networks, Systems Design, Computer Architecture, Software Engineering

EXPERIENCE

Augmented Design Lab — Santa Cruz, California
Research Participant (Advisor: Dr. Jim Whitehead) — Jan 2026 – Mar 2026

- Developed procedural generation prototypes using LLMs (LangChain), Wave Function Collapse, and Z3 constraint solving.
- Implemented constraint-driven content pipelines and grid-based generation systems.
- Analyzed and applied technical research in procedural generation and game systems.

The Audition — Santa Cruz, California
Lead AI & Software Engineer — September 2025 – Current

- Leading software development while coordinating a 13-member multidisciplinary team.
- Designed and trained a PyTorch model to classify guitar solo quality, achieving 85% agreement with human rankings using Spearman rank correlation.
- Architecting and implementing software systems, AI behaviors, and core game infrastructure in Godot.

Integem Inc. — Fremont, California
AI & Robotics Instructor — June 2026 – July 2026

- Taught robotics, computer vision, and generative AI to students ages 10–17.
- Guided students through YOLOv5 dataset creation, annotation, training, evaluation, and hyperparameter tuning.
- Taught deployment of pretrained vision models for robotic object detection and manipulation.
- Mentored students through NVIDIA's Generative AI curriculum, covering diffusion models and applied ML.

The Slug Collective — Santa Cruz, California
Founder & President — Dec 2024 – Current

- Founded and led a 100+ member interdisciplinary music and tech collective at UCSC
- Secured $1,500 in funding through technical project proposals and organizational leadership
- Directed production of a 12-track digital release with 20+ contributors.

PROJECTS

Resonance - Semantic Preset Retrieval (Python, RAG, FastAPI, LLMs) — 2026

- Built a multimodal retrieval-augmented generation (RAG) pipeline combining CLAP audio-text embeddings, vector similarity search, and LLM-guided synthesizer preset refinement.
- Implemented Python + FastAPI backend services for embedding generation, preset retrieval, and parameter manipulation; deployed Dockerized ML pipelines on an Oracle Cloud VM.
- Experimented with LLM-driven iterative preset parameter refinement using natural language feedback.
- Built a Next.js interface enabling real-time semantic search and interactive preset modification

Facial Expression Game Controller (Python, TensorFlow/Keras, OpenCV) — 2026

- Improved CNN validation accuracy by 25.4 percentage points (46.1% → 71.5%) through architecture and training experiments, while keeping the model under 150K parameters.
- Built an OpenCV webcam game controller with face detection, prediction smoothing, confidence gating, and explicit class-to-action mapping.

SmartQueue (Next.js, React, TypeScript, Supabase, Spotify SDK) — 2025

- Built an adaptive queue system that dynamically reorders playlists based on user behavior
- Designed a weighted recommendation algorithm with persistent queue state using Supabase Auth and RLS

SKILLS

Languages: Python, C/C++, TypeScript, SQL, Java, Assembly
AI/ML: PyTorch, TensorFlow/Keras, CUDA, NumPy, SciPy, scikit-learn
Infrastructure: Linux, Docker, PostgreSQL, FastAPI, AWS, Wireshark